

Abdulrahman S. Al-Batati

Robotics Researcher — Autonomous Systems, UAV & Multi-Robot
Logistics, ROS 2 and simulation

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Robotics engineer and researcher with 6+ years developing autonomous ground and aerial platforms end to end, from mechanical design through perception, control, and field deployment. Author of 10+ peer-reviewed publications on GNSS-denied navigation, UAV trajectory prediction, and multi-robot systems, with two patented systems.

Google Scholar: 13 articles, 268 citations, h-index 6, i10-index 6.

RESEARCH FOCUS

Autonomous navigation in GNSS-denied environments	Multi-robot coordination and fleet management	UAV logistics and last-mile aerial delivery
SLAM, VIO, state estimation and sensor fusion	Sim-to-real transfer and large-scale simulation	Learning-based planning and agentic AI for robotics

EXPERIENCE

Senior Research Assistant, Robotics R&D — Prince Sultan University Riyadh · Jan 2022 – Present

- Contributed to the development of a ROS 2-based simulation framework for UAV localization in GPS-denied environments using Gazebo, PX4, and QGC (**PSDSARC project**)
- Evaluated state-of-the-art SLAM and VIO algorithms (ORB-SLAM3, RTAB-Map, Fast-LIO, OpenVINS) for outdoor UAV localization
- Led the development of an autonomous mobile robot (AMIR), including design, control, and deployment (**patented system**)
- Designed and developed RoboEye, a plug-and-play visual-inertial odometry (VIO) sensor system (**patented system**)
- Contributed to a drone-based medical delivery system during Hajj in collaboration with government entities
- Contributed to the development of autonomous UAV and robotic systems across multiple projects, including FASEEH (outdoor autonomous ground vehicle for surveillance) and TAER (VTOL aerial platform)
- Contributed to the development of an autonomous UAV system for aerial target detection and interception (iHunter project)
- Contributed to the Smart PSU Shuttle project, making the campus shuttle autonomous using Autoware and ROS 2
- Developing an autonomous logistics and hospitality delivery robot with local Arabic voice AI, in collaboration with Torod
- Leading two palm tree inspection projects for Red Palm Weevil detection, one using a custom acoustic sensor and one using an e-Nose sensor
- Developing sim-to-real transfer for a Unitree Go2 quadruped to traverse difficult terrain in the real world
- Building a tool to streamline LaTeX authoring for researchers
- Leading the development of an end-to-end generative AI pipeline for automated 3D design and fabrication (natural language → CAD → mesh → G-code → 3D printing)
- Leading research on integrating agentic AI into robotic systems for autonomous decision-making and task execution
- Designed and manufactured 20+ UAV components using 3D printing to improve performance and structural efficiency
- Supervised the development of agricultural drones for spraying and fertilization applications
- Collaborated with international research teams (CTU, CUL, UC Berkeley) on advanced robotics systems

Aircraft Maintenance Engineer — Saudia Aerospace Engineering Industries (SAEI) Jeddah · Aug 2019 – Sep 2021

- Recovered SAR 21M in lost invoices by auditing 23,000+ records and resolving technical discrepancies
- Reduced defect rates by 70% within 4 months by leading a COO-sponsored reliability improvement project
- Identified and resolved 10+ aircraft reliability issues, securing SAR 200K in cost savings through vendor negotiations
- Developed KPI dashboards that reduced analysis time by 96% using advanced Excel solutions
- Represented the department in Aircraft Quality Assurance, ensuring compliance with safety standards

- Designed CAD models and performed CAM/CAE simulations using Siemens NX
- Operated CNC machinery and conducted manufacturing processes in workshop environments
- Analyzed engineering patents and developed structured product architecture reports
- Completed Stratasys training and certification on operating and maintaining the Stratasys J750 3D printer

EDUCATION

M.Sc. Robotics & Autonomous Intelligent Systems — King Fahd University of Petroleum & Minerals, Dhahran (GPA 3.6/4.0, with honor)	Aug 2021
B.Sc. Mechanical Engineering — King Fahd University of Petroleum & Minerals, Dhahran (GPA 3.5/4.0, with honor)	May 2018
Exchange Program, Mechanical Engineering — Colorado School of Mines, Golden, CO, USA	Dec 2016

PUBLICATIONS

1. A Two-Stage Residual Extended Kalman Filter using Extreme Gradient Boosting and Kolmogorov–Arnold Networks for terrain-aided Unmanned Aerial Vehicle Localization in Global Navigation Satellite System-denied environments. **Q1 · IF 9.0** · [10.1016/j.engappai.2026.114430](#)
2. ROS 2 in a Nutshell: A Survey. **First author** · ACM Computing Surveys · **Q1 · IF 30.4** · [10.1145/3815113](#)
3. GNSS-denied unmanned aerial vehicle navigation: analyzing computational complexity, sensor fusion, and localization methodologies. 2nd author (**equal contribution**) · Satellite Navigation · **Q1 · IF 12.4** · [10.1186/s43020-025-00162-z](#)
4. VECTOR: Velocity-Enhanced GRU Neural Network for Real-Time 3D UAV Trajectory Prediction. Co-author · Drones · **Q1 · IF 5.2** · [10.3390/drones9010008](#)
5. Generative 3D Object Modeling for Robust Robot Manipulation in ROS 2. **Presented** at the 2026 International Conference on Advanced Robotics and Mechatronics (ICARM), Paris, France
6. FLIGHTGEN: ROS 2-Powered Automated UAV Dataset Generator. **Presented** at the International Conference on Smart Systems and Emerging Technologies, Marrakech, Morocco · [10.1007/978-3-031-91235-1_28](#)
7. D2DTracker: A Framework for Enabling Real-Time Trajectory Prediction for Agile Drone-to-Drone Tracking via Adaptive Model Selection. **Presented** at the 2024 2nd International Conference on Unmanned Vehicle Systems–Oman (UVS), IEEE, Muscat, Oman · [10.1109/UVS59630.2024.10467173](#)

Full list (Elsevier, IEEE, Springer): [Google Scholar](#)

KEY PROJECTS & PATENTS

- **GPS-Denied UAV Localization** — designing localization systems for high-altitude UAV navigation without GPS
- **RoboEye — VIO Sensor System** — plug-and-play visual-inertial odometry sensor for robotic platforms; patented industrial design, 2026
- **AMIR — Autonomous Mobile Robot** — led full system development from design to deployment; patented industrial design, 2024
- **iHunter — Autonomous UAV Interception** — agile drone system for tracking and intercepting dynamic aerial targets
- **Quadruped RL Locomotion** — reinforcement learning for terrain-adaptive locomotion
- **Agentic AI for Robotics** — LLM-based agents for autonomous planning and execution
- **Generative AI for 3D Design & Fabrication** — end-to-end pipeline converting natural language into manufacturable 3D objects

TECHNICAL SKILLS

Robotics	ROS 2, SLAM, State Estimation, Sensor Fusion, Control Systems
Programming	Python, C++
Simulation	Gazebo, Isaac Sim, MuJoCo, PX4 SITL
AI & Learning	Machine Learning (RNN, LSTM, GRU), Reinforcement Learning (PPO), 3D Generative AI
Computer Vision	OpenCV, Visual Odometry
Design & Manufacturing	CAD, FEA, Design for Additive Manufacturing (DfAM), 3D Printing, Rapid Prototyping
Tools	Git, Docker, LaTeX
Languages	Arabic (native), English (professional), Chinese (beginner, HSK 2)

TRAINING, CERTIFICATIONS & AWARDS

Certifications	Certified 3DEXPERIENCE SOLIDWORKS – Associate; Mechanical Engineer Professional Accreditation, SCE
Professional development	CTU – IEEE RAS Summer School on Multi-robot Systems; IIIA-CSIC – Multi-Agent Reinforcement Learning; University of Alberta – Reinforcement Learning Specialization; Imperial College London – Mathematics for Machine Learning; MIT – Robotics Essentials; MIT – Additive Manufacturing for Innovative Design and Production; The Construct – Managing Fleets of Robots with ROS 2
Awards	1st Place, IEEE RAS Multi-Robot Systems Competition; 2nd Place, Machine Design Championship; 4th Place, Physics University Olympics; supervised winning team, Robotic Sumo Challenge (PSU)

References available upon request.